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Title: Development of a forensically useful age prediction method based on DNA methylation analysis.

Author: Renata Zbieć-Piekarska Magdalena Spólnicka Tomasz Kupiec Agnieszka Parys-Proszek Żanetta Makowska Anna Pałeczka Krzysztof Kucharczyk Rafał Płoski Wojciech Branicki

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Title page

Development of a forensically useful age prediction method based on DNA methylation analysis.

Renata Zbieć – Piekarska[1], Magdalena Spólnicka[1], Tomasz Kupiec[2], Agnieszka Parys-Proszek[2] , Żanetta Makowska[1], Anna Pałeczka[1], Krzysztof Kucharczyk[3], Rafał Płoski[4], Wojciech Branicki[2,5]*

[1] – Central Forensic Laboratory of the Police, Warsaw, Poland

[2] – Institute of Forensic Research, Krakow, Poland

[3] – BioVectis, Warsaw, Poland

[4] – Warsaw Medical University, Warsaw, Poland

[5] – Faculty of Biochemistry, Biophysics and Biotechnology, Jagiellonian University, Krakow, Poland

* - corresponding author

Wojciech Branicki

Institute of Forensic Research

Westerplatte st. 9; 31-033 Kraków, Poland

e-mail: wbranicki@ies.krakow.pl

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[5] – Faculty of Biochemistry, Biophysics and Biotechnology, Jagiellonian University, Krakow, Poland

* - corresponding author
Wojciech Branicki
Institute of Forensic Research
Westerplatte st. 9; 31-033 Kraków, Poland
e-mail: wbranicki@ies.krakow.pl

Highlights

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Abstract

Forensic DNA phenotyping needs to be supplemented with age prediction to become a relevant source of information on human appearance. Recent progress in analysis of the human methylome has enabled selection of multiple candidate loci showing linear correlation with chronological age. Practical application in forensic science depends on successful validation of these potential age predictors. In this study, eight DNA methylation candidate loci were analysed using convenient and reliable pyrosequencing technology. A total number of 41 CpG sites was investigated in 420 samples collected from men and women aged from 2 to 75 years. The study confirmed correlation of all the investigated markers with human age. The five most significantly correlated CpG sites in *ELOVL2* on 6p24.2, *C1orf132* on 1q32.2, *TRIM59* on 3q25.33, *KLF14* on 7q32.3 and *FHL2* on 2q12.2 were chosen to build a prediction model. This restriction allowed the technical analysis to be simplified without lowering the prediction accuracy significantly. Model parameters for a discovery set of 300 samples were $R^2 = 0.94$ and the standard error of the estimate = 4.5 years. An independent set of 120 samples was used to test the model performance. Mean absolute deviation for this

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testing set was 3.9 years. The number of correct predictions ±5 years achieved a very high level of 86.7% in the age category 2-19 and gradually decreased to 50% in the age category 60-75. The prediction model was deterministic for individuals belonging to these two extreme age categories. The developed method was implemented in a freely available online age prediction calculator.

Keywords

DNA methylation, prediction modelling, DNA-based age prediction, forensic science

1. Introduction

Forensic DNA phenotyping (FDP) is having a great impact on contemporary forensic genetics. By extending information that can be retrieved from DNA, FDP is creating a new dimension of DNA-based intelligence offered by forensic science. Prediction of blue and brown eye colours, red hair colour and extreme skin colours can be achieved with high accuracy and involving a very limited number of DNA markers [1], [2], [3], [4], [5], [6], [7]. The genetics of other externally visible characteristics (EVCs) like human height or facial morphology seem to be much more complex and their accurate prediction may involve analysis of thousands of polymorphisms [8], [9]. It is, however, important to note that progressing age changes the appearance of an individual remarkably and thus age estimation will be necessary to make a DNA-based sketch of an unknown suspect. Aging has a clear impact on most EVCs but may more seriously affect some like hair colour or male pattern baldness. This makes the development of a reliable method of age prediction a highly sought-after capability in the field of forensic genetics. Recent progress in the field of DNA methylation has enabled selection of multiple candidate loci showing linear correlation of methylation status with biological age [10], [11], [12], [13]. The locus *ELOVL2* is the most thoroughly evaluated marker of age so far [12], [14], [15], [16]. It has been shown that this locus can be reliably analysed in old bloodstains, which are a main source of DNA in forensic laboratories [17]. Importantly, *ELOVL2* has been shown to be correlated with age in various human tissues [12]. Although *ELOVL2* can be considered as a very powerful age predictor,

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this marker needs to be enhanced by additional markers to improve the overall accuracy of age estimation. Further studies may continue to search for novel DNA methylation markers or focus on already known candidate loci. It is, however, commonly agreed that replication is a significant aspect of validation of any new marker that is of interest to forensic science. Moreover, the special characteristics of DNA methylation necessitate thorough validation of the selected age predictors in various human populations and tissue types. The first age prediction models based on a small number of powerful predictors have recently been reported to estimate age with reasonable accuracy [11], [13]. Interestingly, the *ELOVL2* marker was among the predictors in an age prediction model involving 71 DNA methylation loci that can predict chronological age in blood with a correlation coefficient of 96% and standard error of 3.9 years [12]. This model, although highly accurate, may not be practical to implement in forensic science due to the high number of DNA methylation loci that need to be examined, which is a limiting factor for their routine use in high-throughput laboratories. Therefore in this work we further investigated the eight most promising loci reported in Hannum et al. using a highly reliable, cost effective and easy to apply pyrosequencing method. The number of CpG sites investigated was extended to a final number of 41 CpGs subjected to analysis. The obtained results allowed us to select the 5 most informative CpG sites and develop an age prediction method that combines simplicity of technical analysis with high accuracy of prediction. The model was evaluated on an independent set of samples and a detailed investigation of prediction accuracy was performed in four defined age categories. Finally, the method was implemented in a simple age calculator that is freely available online.

2. Materials and Methods

2.1. The study population

The study was approved by the commission of bioethics at the Institute of Cardiology in Warsaw, no IK-NP-0021-79/1396/13. All participants signed informed consent statements prior to sample donation. Peripheral blood was collected in EDTA containing tubes from 427 unrelated males and females aged between 18 and 75 years. Additionally, samples from children aged 2-17 years were collected and written consents were obtained from their parents.

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2.2. Analysis of DNA methylation in 41 CpG sites from eight candidate loci

Eight candidate loci were selected from Hannum et al., based on evaluation of DNA methylation results provided in supplementary materials [12]. Linear regression coefficients (β) and significance of correlation were used to make a distinction. Characteristics of the selected loci are shown in Table 1. Table 1 also gives information about CpG sites studied in Hannum et al. Total DNA was extracted from the obtained blood samples as previously described [17]. Results of DNA methylation analysis in 7 CpG sites located in the *ELOVL2* promoter were previously reported [17]. Seven pyrosequencing assays for *C1orf132*, *KLF14*, *SLC6A4*, *F5*, *CCDC102B*, *TRIM59* and *FHL2* were designed using the PyroMark Assay Design 2.0 (Qiagen, Hilden, Germany). The Qiagen 96-well bisulfite conversion kit and manufacturer's protocol were used to convert unmethylated cytosines to uracils [Qiagen, Hilden, Germany]. 2 µg of DNA was subjected to bisulfite conversion. PCR and sequencing primer sequences, their concentrations and the sequences to be analysed are given in supplementary Table S1. PCR reactions were carried out in a total volume of 25 µL, containing 0.2 µM of each of the primers, 20 ng of template DNA, and PyroMark PCR Master Mix, containing HotStarTaq DNA Polymerase, 1x PyroMark PCR Buffer and dNTPs. The amplification program consisted of an initial denaturation step at 95°C for 10 min, followed by 40-45 cycles comprising: denaturation (94°C for 30 s), annealing (54-56°C for 30 s) and extension (72°C for 30 s), and a final extension of 72°C for 10 min. Negative PCR controls were included in each PCR amplification. Pyrosequencing was performed using Pyro Gold reagents on a PyroMark vacuum prep workstation and a PyroMark Q24 instrument, following the manufacturer's instructions. The generated pyrograms were automatically analyzed using PyroMark analysis software (Qiagen, Hilden, Germany).

2.3. Statistic analysis

The correlation between age and the methylation level at the 41 investigated CpGs was analyzed in a discovery set of 300 subjects ranging in age from 2 to 75 years. Standardized regression coefficients (β) were used to compare the effect size of particular loci. Multivariate linear regression enabled simultaneous analysis of all the tested CpG sites. The prediction accuracy associated with individual CpGs and the developed prediction model was assessed using the adjusted R^2 parameter, which is a measure of the proportion of age variance explained by particular predictors and their combined effect. The same DNA methylation data obtained for 300 blood samples were then used to build a linear regression age prediction model. The final model was developed based on results of multivariate regression analysis,

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restricting the number of involved CpGs to the 5 best predictors. The model was further validated using an additional set of 120 samples. The mean absolute deviation (MAD) from the chronological age was calculated for the entire model building and model testing sets, for subsets of the four distinct age categories: 1 – individuals at age 2-19; 2 – individuals at age 20-39; 3 – individuals at age 40-59 and 4 – individuals at age 60-75 as well as for females and males separately. Additionally, each obtained result was interpreted as either correct if the predicted age was concordant with the chronological age ± 5 years, or incorrect if these values differed by more than ± 5 years. All the analyses were performed using IBM SPSS statistics ver. 22.

2.4 The linear regression equation and age prediction calculator

The equation corresponding to the developed age prediction model based on 300 discovery samples and implemented in the online age prediction calculator was as follows:

Predicted age (years) = 3.26847784751817 + 0.465445549010653×methC7-ELOVL2 – 0.355450171437202×methC1-C1orf132 + 0.306488541137007×methC7-TRIM59 + 0.832684435238792×methC1-KLF14 + 0.237081243617191×methC2-FHL2

The calculator is freely available at www.agecalculator.ies.krakow.pl.

3. Results

3.1 Development of the age prediction model

Age prediction modelling was based on DNA methylation data for 41 CpGs in eight DNA methylation loci obtained using a method based on bisulfite conversion followed by pyrosequencing. DNA methylation data for 7 CpGs in *ELOVL2* were retrieved from our previous study involving 427 peripheral blood samples (307 included in a discovery set) collected from individuals aged from 2 to 75 years [17]. The same subjects were included in analysis of 3 CpGs in *C1orf132*, 4 CpGs in *KLF14*, 4 CpGs in *SLC6A4*, 3 CpGs in *F5*, 2 CpGs in *CCDC102B*, 8 CpGs in *TRIM59* and 10 CpGs in *FHL2*. Analysis of seven samples failed in a number of loci and these samples were thus excluded from further statistical analyses. The lowest sequencing completion rate of 98% was noted for the locus *CCDC102b*. For the remaining loci the sequencing completion rate ranged from 98.3% to 100%. The DNA methylation status in the 300 discovery samples was used to test for correlation between the analysed CpGs and chronological age. Supplementary Table S2 contains data concerning the effect size of the studied CpGs as measured by standardized linear regression coefficients (β).

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Position C7 in *ELOVL2* (R=0.912) showed the strongest correlation, explaining 83% of age variance in this sample set. Position C1 in *C1orf132* was the second most strongly correlated site. This CpG had an R value of 0.889, explaining 79% of age variance. The third best single age predictor was position C5 in *TRIM59* with R=0.863 and explained age variance of 74%. Multivariate regression analysis revealed that 10 CpG sites in all but the *CCDC102B* locus provided independent information concerning human chronological age in this sample set (Supplementary Table S2). Such a model explained 95.3% of age variance with an average error of 4.0 years. Aiming to develop a method that would combine technical simplicity and accuracy of age estimation, we decided to restrict the number of analysed loci to the five most informative predictors in the model, which included (in order of their predictive power): C7 in *ELOVL2*, C1 in *C1orf132*, C7 in *TRIM59*, C1 in *KLF14* and C2 in *FHL2* (Table 2). This age prediction model had only slightly worse parameters, explaining 94.1% of age variance with a standard error of 4.5 years. The mean absolute deviation (MAD) calculated for the entire discovery set of 300 samples was 3.4 years. Detailed analysis of MAD in four age categories (2-19, 20-39, 40-59 and 60-75) clearly indicated that MAD rises with subjects' chronological age. The lowest MAD value (2.7) was calculated for category 1 and the highest value (5.0) for category 4 (Table 3). The MAD value was also lower for women (3.0) than men (3.7). Gender was found to slightly contribute to the aging rate in our study but this effect was weak (R^2 improved by 0.001 with age as a covariate) and thus we decided not to include gender in prediction modelling. Figure 1 presents a plot with chronological age versus predicted age (R^2=0.943). DNA methylation data obtained for 41 CpG sites studied in the discovery set of 300 samples are gathered in Supplementary Table S3.

3.2 Testing of the age prediction model

The testing set included 120 independent samples from 63 women and 57 men aged from 2 to 75 years and the value of MAD for this sample set was 3.9 years. MAD also increased with age and for the youngest subjects (category 1) was 2.8 years, while for the oldest subjects (category 4) was 5.6 years (Table 3). The effect of gender was also noticeable in the testing set and thus MAD in females was 3.7, while in males it equalled 4.0. Based on the assumption that age prediction was correct when predicted and actual ages differed max ±5 years, a total number of 72% correct predictions was observed in the set of 120 testing samples. Supplementary Figure 1 shows that dispersion of absolute deviation from chronological age was higher in the case of age categories 3 and 4, where deviation for some samples exceeded 10 years. In concordance with the difference in MAD values observed in particular age

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categories, the number of correct predictions decreased with (increasing) age category (Table 3). The highest number of correct predictions (86.7%) was observed in the category including the youngest subjects (2-19), while the lowest number of correct predictions (50%) was observed in the category including the oldest subjects (60-75). Figure 2 shows that the developed age prediction model was able to unambiguously distinguish the two extreme age categories 1 and 4, comprising youngest and oldest subjects. Figure 3 presents a plot with chronological age versus predicted age of the testing set (R^2=0.944). DNA methylation data obtained for five loci studied in the testing set of 120 samples are summarized in Supplementary Table S4. The developed age prediction equation was implemented in a simple calculator that is available online at the webpage: http://www.agecalculator.ies.krakow.pl.

4. Discussion

Forensic science has for a long time been seeking a relevant type of marker facilitating age prediction from biological traces associated with crime scenes [18]. Although sjTREC molecules decreasing with age in humans in relation to thymus involution constitute a reasonable marker for human age estimation [19], it seems that DNA methylation shall now be considered as the most promising information source about human age in forensic science. Not only because the study of DNA methylation is not restricted to blood, but because of the high abundance of DNA methylation loci in the human genome which are linearly correlated with age and can complement each other, providing a very accurate final estimate of chronological age. Eight highly age-correlated DNA methylation loci were selected based on preliminary analysis of DNA methylation data provided in Hannum et al. Standardized regression coefficients were calculated for 71 age-correlated markers in a population of 656 individuals differing in age from 19 to 101 years reported in the above mentioned paper [12]. These eight loci were further validated on an independent set of 300 blood samples used in our study as a training set and the five best predictors were included in a prediction model. The developed model includes *ELOVL2*, *FHL2*, *KLF14*, and *TRIM59*, where methylation increases with age, and *C1orf132*, where methylation decreases with age.

The gene *ELOVL fatty acid elongase 2 (ELOVL2)* is located on 6p24.2 and encodes for a 296–amino acid transmembrane enzyme protein involved in the synthesis of very long chain polyunsaturated fatty acids (PUFA). Correlation of DNA methylation status in the *ELOVL2* promoter with chronological age has been very well documented [12], [14], [15],

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[16], [17]. It has also been confirmed that this locus can be analysed in old bloodstains, which makes it a useful marker in forensic investigations [17].

The gene *FHL2* on 2q12.2 encodes for a 279–amino acid protein, which is a member of four-and-a-half-LIM domain-containing proteins. *FHL2* has been shown to be involved in interaction with the androgene receptor gene (AR) [20], [21]. In particular, Muller et al. (2000) postulated that *FHL2* increased the transcriptional activity of the AR. It is interesting to note that some AR alleles significantly increase susceptibility to male pattern baldness (MPB) [22]. Prediction of male pattern baldness is of interest to forensic science and since MPB is a progressive condition, age represents an important factor influencing interpretation of the MPB risk genotypes.

The gene *KLF14* on 7q32.3 is intronless and encodes for a 323–amino acid protein belonging to the Kruppel-like family of transcription factors. It has been shown that *KLF14* is maternally-expressed and is associated with susceptibility to metabolic diseases like Type 2 Diabetes and HDL-cholesterol levels [23].

The gene *TRIM59* on 3q25.33 encodes a 403–amino acid protein, which may be involved in regulation of immune signalling pathways [24]. Interestingly, *TRIM58*, which is a member of the same very conservative protein superfamily, has been correlated with age in a previous study [10]. It has been noted that many TRIM family members are induced by interferons involved in resistance and response to pathogens [25].

Research explaining the function and role of chromosome 1 open reading frame 132 (*C1orf132*) located on 1q32.2 may be very exciting since this locus seems to be important in human aging. Correlation of position cg10501210 with age in Hannum et al. was high, with R^2=0.526, and in our study this parameter, obtained for a neighbouring position C1, was even stronger (R^2=0.789). Interestingly, expression of a protein encoded by *C1orf132* could be expected to increase with aging according to the progressing DNA hypomethylation of this locus. The statistical parameters locate *C1orf132* very high on a list of human age predictors.

In addition to *C1orf132*, we also observed higher values of R^2 compared to Hannum et al. for the remaining loci included in the age prediction model. The best age predictor, *ELOVL2*, in our study had R^2=0.832 (position C7), while in Hannum et al. cg16867657=0.737. The *FHL2* locus had R^2 for the position C1=0.712, while in Hannum et al., cg06639320 had R^2=0.557. R^2 for the *TRIM59* locus in our study was 0.690 (position C7), while in Hannum et al., cg07553761=0.489. Finally, *KLF14* in our study had R^2=0.535, while in Hannum et al. this parameter calculated for cg14361627 was 0.376. Supplementary Figures 2–6 present plots with chronological age versus predicted age for the five CpGs included in

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the final prediction model. We can speculate that higher R^2 values in our study may be related to the higher number of younger individuals, for which age is estimated with lower error as will be discussed further. We checked that among 656 subjects investigated in Hannum et al., only one individual belonged to our age category 1 (2-19), and category 2 (20-39) was also underrepresented with only 5.2% of the overall number of subjects, while individuals from our category 4 (60-75 years) and older comprised 62.3% of the entire set.

Our results obtained for both model-building and model-testing sets are concordant, showing that age prediction accuracy decreases with actual age and is lowest in the age category comprising individuals aged 60-75 years. In our data, the absolute differences between predicted and actual ages are larger in the categories of older people. The problem of a decrease in prediction accuracy with age is reflected in MAD values. MAD=5.6 was obtained for samples included in the 4th age category of the testing set and the number of correct predictions ±5 years in this category was merely 50%. The same trend of decreasing prediction accuracy with age was observed for our two best predictors – positively correlated *ELOVL2* [17] and negatively correlated C1orf132. In the case of a prediction model based on the latter, MAD increased with age category – from 4.2 years in category 1 to 12.8 years in category 4 (across the entire dataset, MAD=7.0, data not shown). More accurate age prediction in adolescents and children has also been reported by Horvath [26]. It is worth noting that progressing age changes physical appearance most significantly in younger individuals, where age prediction using our model is most relevant (78.3% correct predictions ±5 years were noted for subjects aged 2-59 years). Studies on human methylome have reported many loci which show linear correlation with age, but it has also been shown that DNA methylation in some loci may be a subject of rapid changes [27]. It seems to be clear that environmental factors may cause variation in methylome irrespective of the genetic constitution of an organism. It is also becoming clear that DNA methylation status reflects biological rather than chronological age. Moreover, our data show that age prediction in older people is more prone to errors. Recently, it has been shown that the difference between age indicated by DNA methylation and chronological age is an independent predictor of mortality risk [28]. It has been concluded that DNA methylation estimated age can be considered as an "epigenetic clock", which in fact reflects biological age, which in many cases runs parallel with chronological age [26], [28].

In some cases, CpGs selected in our model were not the same as those present on the Illumina Infinium HumanMethylation450 BeadChip used in the Hannum et al. study. Further studies are necessary to explain these differences. The constrained limitation of the number of

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predictors included in the model to only five loci makes technical analysis achievable in a typical forensic laboratory. It is also worth noting that pyrosequencing has potential for multiplexing, which can further simplify the protocol and reduce the cost of technical analysis [29].

Our 5-CpGs age prediction model explained 94% of age variance with an error of 4.5 years. Examination of all 71 predictors included in the model reported by Hannum et al. provided slightly higher accuracy of prediction as measured by a standard error of 3.9 years. Apparently, by increasing the number of DNA methylation loci we can expect better accuracy of age prediction. This is also clear from other studies like one showing a prediction model based on 100 CpGs that estimated age with a standard error of 4.3 years [13] or a study reporting a prediction model employing 353 markers and estimating age with a median absolute deviation of 3.6 years [26]. However, the loss of accuracy observed for prediction models based on a restricted number of predictors seems not to be substantial [13] [11].

Notably, our model not only estimates human age with very good accuracy over the age range of 2-75 years, but is also deterministic in the differentiation of children/teenagers (2-19 years) from individuals of advanced age (≥60 years). If confirmed, the test would have a similar value when predicting young vs. old individuals to the widely used amelogenin gender test [30].

In our study, gender had a noticeable but very small effect on age prediction. Although the effect of gender on age prediction has not been detected in all studies [10], our results are concordant with Weidner et al. (2014), who found that the predicted age was higher in men than in women. Also MAD for women included in our testing set was 3.7 years, while for men this parameter was 4.0 years (data not shown). It has previously been estimated that DNA methylation in males changes 4% faster than in women [12].

An interesting problem, which needs further study, is the tissue specificity of DNA methylation patterns. The available literature indicates that the differences in DNA methylation characteristic for various tissues may influence age prediction but some markers like *ELOVL2* may remain relatively stable predictors [12]. In this context, a study by Koch and Wagner, who analysed DNA methylation data for various tissues, is very interesting [10]. Average MAD for various tissue types was 11.4 years, but the authors noticed that for some sample sets MAD was lower than 6 years. The correlation coefficients calculated for particular loci also differed a lot in various tissues. The same conclusion can be drawn from the study by Horvath, who calculated median absolute difference between actual age and predicted age for various tissues and cell types. Differences in error were substantial for

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various tissues. For some sample sets, the error was very small (CD4 T cells from new-borns – 0.27, buccal epithelium – 0.37), for some it was acceptable (saliva – 2.7, whole blood – 3.7, kidney – 4.6, cerebellar samples – 5.9, liver – 6.7), while for some it was very high (heart – 12, muscle – 18) [26]. These studies confirmed that finding universal markers providing the same age prediction accuracy in various tissues might be difficult. Selection of such a set of markers that are linearly correlated with age across tissues and body fluids would be most appropriate in forensic science, but further studies may show if this ambitious task is realistic. It is also worth noting that previous studies have shown tumour cells age more quickly compared to normal cells, adding another aspect of human age prediction that needs further investigation from the perspective of forensic science [12].

5. Conclusions

Analysis of DNA methylation in 5 CpG sites located in *ELOVL2, C1orf132, TRIM59, KLF14* and *FHL2* enabled estimation of human chronological age with high accuracy as measured by MAD=3.9 for the testing set. 78.3% correct predictions ±5 years were noted in subjects aged 2-59 years. This number was only 50% in the age category comprising individuals aged 60-75 years, indicating that DNA methylation is less efficient in age predicting in older individuals. This result may be explained by environmental factors contributing more significantly to the DNA methylation status in older individuals whose medical history or life style differences increase age estimation error. The developed equation implemented in a convenient age calculator together with the optimised pyrosequencing assays provide a relevant tool for forensic age prediction from blood samples. Further studies are needed to assess the usefulness of the proposed method for estimating age from other specimens like saliva or semen. The complete training and testing datasets enclosed to this manuscript can be further used to continue works on development of the most accurate and convenient forensic age prediction system based on DNA methylation markers.

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Description of Figures, Tables and Supplementary materials

Figure 1. Predicted versus chronological age of the discovery set of 300 samples. Prediction made using the model including positions C7 in *ELOVL2*, C1 in *C1orf132*, C7 in *TRIM59*, C1 in *KLF14* and C2 in *FHL2*. Negative prediction values obtained for the 6 youngest individuals, 2 and 3 years old, were set at 0. $R^2 = 0.943$.

Figure 2. Predicted age in four age categories. 1 – age category 2-19; 2 – age category 20-39; 3 – age category 40-59; 4 – age category 60-75.

Figure 3. Predicted versus chronological age of the testing set of 120 samples. Prediction made using the model including positions C7 in *ELOVL2*, C1 in *C1orf132*, C7 in *TRIM59*, C1 in *KLF14* and C2 in *FHL2*. Negative prediction value obtained for the 1 youngest individual, 2 years old, was set at 0. $R^2 = 0.944$.

Table 1. Characteristics of the analyzed age related candidate DNA methylation loci.

Table 2. Final age prediction model including five best age predictors: ELOVL2, C1orf132, TRIM59, KLF14 and FHL2.

Table 3. MAD and percentage of correct predictions in four age categories.
1 – age category 2-19; 2 – age category 20-39; 3 – age category 40-59; 4 – age category 60-75. Correct prediction was assumed when predicted and actual age matched ±5 years.
Supplementary Table S1. PCR and sequencing primer sequences, their concentrations and sequences to analyse.

Supplementary Table S2. Univariate and multivariate regression analysis of the 41 CpG sites in eight DNA methylation loci.

Supplementary Table S3. DNA methylation values obtained for the 300 sample discovery set in the 41 analysed CpG sites located in eight markers included in the study.

Suplementary Table S4. DNA methylation values obtained for the 120 sample testing set in the 32 analysed CpG sites located in five markers included in the model.

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Supplementary Figure S1. Absolute difference between actual and predicted age in four age categories. 1 – age category 2-19; 2 – age category 20-39; 3 – age category 40-59; 4 – age category 60-75.

Supplementary figure S2. Predicted versus chronological age of the discovery set of 300 samples. Prediction made using the position C7 in *ELOVL2*. Negative prediction values obtained for the 6 youngest individuals, 2 and 3 years old, were set at 0. $R^2 = 0.848$.

Supplementary figure S3. Predicted versus chronological age of the discovery set of 300 samples. Prediction made using the position C1 in *C1orf132*. $R^2 = 0.79$.

Supplementary figure S4. Predicted versus chronological age of the discovery set of 300 samples. Prediction made using the position C7 in *TRIM59*. $R^2 = 0.691$.

Supplementary figure S5. Predicted versus chronological age of the discovery set of 300 samples. Prediction made using the position C1 in *KLF14*. $R^2 = 0.537$.

Supplementary figure S6. Predicted versus chronological age of the discovery set of 300 samples. Prediction made using the position C2 in *FHL2*. Negative prediction values obtained for the 2 youngest individuals, 2 years old, were set at 0. $R^2 = 0.711$.

Table 1. Characteristics of the analyzed age related candidate DNA methylation loci.

Locus	Gene function	CpG site	Position in Hannum et al., 2013	Chromosome location (GRCh38)
ELOVL2	synthesis of very long chain polyunsaturated fatty acids	C1	cg21572722	chr6: 11044894
		C2	cg24724428	chr6: 11044888
		C3	-	chr6: 11044880
		C4	cg16867657	chr6: 11044877
		C5	-	chr6: 11044875
		C6	-	chr6: 11044873
		C7	-	chr6: 11044867
C1orf132	unknown	C1	-	chr1:207823681
		C2	cg10501210	chr1:207823675
		C3	-	chr1:207823672
KLF14	transcription factor	C1	cg14361627	chr7:130734355
		C2	-	chr7:130734357
		C3	-	chr7:130734373
		C4	-	chr7:130734375
SLC6A4*	serotonine transporter	C1	-	chr17:30235578
		C2	-	chr17:30235586
		C3	-	chr17:30235591
		C4	-	chr17:30235604
F5	factor involved in blood coagulation	C1	-	chr1:169586788
		C2	cg16054275	chr1:169586784

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		C3	-	chr1:169586774
CCDC102B	unknown	C1	cg19283806	ch18:68722183
		C2	-	ch18:68722210
TRIM59	regulator of immune signaling pathways	C1	-	chr3: 160450172
		C2	-	chr3: 160450174
		C3	-	chr3: 160450179
		C4	-	chr3: 160450184
		C5	cg07553761	chr3: 160450189
		C6	-	chr3: 160450192
		C7	-	chr3: 160450199
		C8	-	chr3: 160450202
FHL2	transcription factor	C1	cg06639320	chr2: 105399282
		C2	-	chr2: 105399288
		C3	-	chr2: 105399291
		C4	-	chr2: 105399297
		C5	-	chr2: 105399300
		C6	cg22454769	chr2: 105399310
		C7	cg24079702	chr2: 105399314
		C8	-	chr2: 105399316
		C9	-	chr2: 105399323
		C10	-	chr2: 105399327

* - in *SLC6A4* Hannum et al. studied cg14692377, which was not investigated in our study

Table 2. Final age prediction model including five best age predictors: ELOVL2, C1orf132, TRIM59, KLF14 and FHL2.

Locus	CpG site	Location (GRCh38)	R	Corrected R^2	Standard error	P-value
ELOVL2	C7	Chr6:11044661	0.971	0.941	4.457	1.1121×10^{-23}
C1orf132	C1	Chr1:207823681				3.1724×10^{-23}
TRIM59	C7	Chr3: 160450199				3.3826×10^{-12}
KLF14	C1	Chr7:130734355				4.353×10^{-12}
FHL2	C2	Chr2: 105399288				2.3384×10^{-8}

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Table 3. MAD and percentage of correct predictions in four age categories.

1 – age category 2-19; 2 – age category 20-39; 3 – age category 40-59; 4 – age category 60-75. Correct prediction was assumed when predicted and actual ages matched ±5 years.

	Training set = 300		Testing set = 120	
Age category	MAD	Correct predictions (%)	MAD	Correct predictions (%)
1	2.7	86.7	2.8	86.7
2	2.9	85.7	3.3	76.7
3	3.4	76.2	3.9	71.9
4	5.0	55.2	5.6	50.0
Overall	3.4	77.1	3.9	71.7

Supplementary Table S3. DNA methylation values obtained for the 300 sample discovery set in the 41 analys
CpG sites located in eight markers included in the study.

sampl e	age	sex	sex_bi n	c1orf1 32_c1	c1orf1 32_c2	c1orf1 32_c3	klf14_c 1	klf14_c 2	klf14_c 3	klf14_c 4	slc6a4 _c1	slc6a4 _c2	slc6a4 _c3	slc6a4 _c4	
1	19	female	0	86	68	72	5	2	7	2	3	6	8	30	
2	16	male	1	89	70	77	4	2	7	1	2	3	12	21	
3	19	female	0	89	76	80	4	2	6	2	2	6	16	15	
4	13	female	0	91	78	82	3	2	6	1	2	5	12	19	
5	18	male	1	89	75	79	7	3	8	2	2	4	9	22	
6	18	female	0	81	62	67	5	2	8	1	2	5	11	29	
7	19	female	0	86	69	74	4	2	7	1	2	7	12	34	
8	19	female	0	90	74	80	3	1	7	1	2	5	8	19	
9	18	female	0	79	59	62	4	2	8	1	3	7	13	21	
10	19	female	0	86	64	71	3	1	5	1	1	4	10	15	
11	19	female	0	84	71	72	4	2	8	2	3	6	13	22	
12	18	female	0	87	72	76	5	3	9	1	2	6	17	31	
13	19	male	1	89	69	76	5	2	8	2	2	4	8	31	
14	18	male	1	91	76	80	4	3	7	2	2	6	12	24	
15	16	female	0	88	71	75	4	2	8	1	2	5	14	17	
16	17	male	1	89	72	78	4	2	6	1	2	7	17	21	
17	17	male	1	90	75	79	3	2	6	1	2	5	11	16	
19	19	male	1	88	71	75	6	3	8	2	2	4	6	23	
20	18	male	1	90	76	79	5	3	7	2	2	7	11	20	
21	11	male	1	92	82	85	5	4	8	3	2	6	9	22	
26	59	male	1	54	38	39	10	5	12	3	2	6	16	17	
27	57	male	1	71	53	55	8	4	10	3	2	6	10	40	
28	53	male	1	60	46	47	7	4	9	2	2	5	4	10	
29	52	male	1	64	48	48	10	6	12	3	2	7	29	20	
30	55	male	1	46	31	31	10	6	11	4	2	5	15	17	
31	52	male	1	78	60	62	13	6	12	4	2	6	15	16	
32	55	female	0	60	43	43	9	5	10	3	2	5	15	20	
33	51	female	0	46	35	36	11	6	11	4	2	6	15	20	
34	53	male	1	72	52	54	8	4	12	3	3	7	17	22	
35	54	male	1	58	41	42	12	6	12	4	2	5	14	18	

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36	50 male	1	59	42	41	9	5	9	3	2	7	17	17
38	53 male	1	54	35	36	11	7	14	5	3	5	12	15
39	52 male	1	70	52	52	11	6	12	4	2	7	16	19
40	50 male	1	59	40	41	9	5	11	3	2	4	12	15
1_A3	14 male	1	88	75	79	3	2	6	1	2	3	12	10
1_A5	22 male	1	79	61	65	4	1	5	1	2	4	12	13
1_A11	27 female	0	78	60	62	8	5	8	3	1	4	14	17
1_A12	24 female	0	77	58	62	4	2	6	1	2	3	13	14
1_B3	25 male	1	87	72	78	6	3	7	1	2	6	19	13
1_B4	27 female	0	82	57	64	5	3	8	2	1	4	13	15
1_B5	20 female	0	83	62	70	4	2	5	1	1	4	14	14
1_B6	20 female	0	84	69	72	3	2	6	2	2	5	18	18
1_B8	29 female	0	82	65	66	8	3	8	2	2	5	14	17
1_B9	25 male	1	79	59	64	5	2	5	2	2	5	13	15
1_B10	20 female	0	81	62	66	4	2	6	1	5	8	19	19
1_B11	28 male	1	76	60	63	6	3	7	2	1	5	13	15
1_B12	29 female	0	67	48	49	5	3	7	1	1	5	16	19
1_C1	25 female	0	78	62	66	4	2	7	1	3	4	14	17
1_C3	28 male	1	77	59	62	2	0	4	1	2	6	11	15
1_C5	30 male	1	76	55	58	5	1	7	2	2	3	15	16
1_C6	38 male	1	70	59	56	4	2	7	1	2	5	15	16
1_C10	35 male	1	75	53	57	8	4	5	1	2	5	13	15
1_C11	36 male	1	70	47	52	8	4	11	3	3	5	12	15
1_C12	31 female	0	73	50	52	4	2	8	3	2	5	13	15
1_D1	31 male	1	73	55	57	4	2	6	1	1	4	14	16
1_D2	33 female	0	74	56	61	6	3	6	1	2	5	14	17
1_D3	30 male	1	73	53	58	7	3	6	2	2	4	12	15
1_D4	30 female	0	78	64	67	5	2	7	2	2	6	17	18
1_D5	37 male	1	64	45	48	5	2	6	1	2	5	16	18
1_D7	33 female	0	70	55	55	5	2	7	2	2	5	14	20
1_D8	38 female	0	74	50	54	6	4	8	0	2	4	13	15
1_D10	36 male	1	80	64	66	7	3	8	2	2	7	14	16
1_D11	31 male	1	75	55	59	8	3	7	3	2	4	14	15
1_D12	31 female	0	66	50	52	5	4	8	3	2	5	19	18
1_E1	32 male	1	81	57	62	5	2	6	2	1	4	12	14
1_E2	36 male	1	72	48	51	8	3	7	3	2	5	13	15
1_E3	39 male	1	66	44	45	7	3	8	2	2	4	15	15
1_E4	35 male	1	71	54	58	5	2	6	2	2	4	13	20
1_E5	39 female	0	68	49	55	5	2	7	2	2	5	16	18
1_E7	42 male	1	71	57	59	10	4	9	3	2	7	14	20
1_E8	40 female	0	64	44	48	6	4	5	0	2	6	18	25
1_E9	44 male	1	63	40	43	6	5	9	3	1	5	16	15
1_E10	44 female	0	67	49	52	8	5	9	3	2	7	19	19
1_E11	41 female	0	70	45	46	7	3	7	2	1	4	12	13
1_E12	40 female	0	68	48	50	6	3	7	2	3	5	18	18
1_F1	43 female	0	51	36	38	5	3	7	2	2	6	16	16
1_F2	48 male	1	65	46	51	6	4	8	2	2	6	16	21
1_F3	47 male	1	46	30	29	7	3	8	2	1	4	12	14
1_F5	41 male	1	67	49	53	7	4	9	2	3	7	24	25
1_F6	44 female	0	56	40	42	4	1	8	3	1	3	12	19
1_F7	43 male	1	57	36	39	6	3	7	2	2	5	16	17
1_F8	44 female	0	65	46	49	10	7	13	7	1	9	18	16
1_F9	46 male	1	61	39	41	6	3	8	2	2	5	14	20
1_F10	46 female	0	64	44	47	10	6	9	3	2	5	18	20
1_F11	41 female	0	63	43	46	6	2	7	2	1	4	11	14

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1_F12	43 male	1	69	47	49	7	6	8	4	2	6	10	15
1_G1	41 male	1	65	49	53	7	3	7	2	2	5	15	17
1_G2	53 female	0	40	30	31	6	3	8	2	2	5	14	19
1_G4	53 female	0	51	34	39	7	4	7	3	3	7	26	23
1_G6	60 male	1	52	36	36	11	8	7	3	7	7	18	24
1_G7	61 male	1	44	33	32	12	6	9	3	0	4	12	12
1_G8	72 male	1	44	31	32	18	13	17	5	3	6	14	18
1_G9	63 male	1	48	34	33	11	4	8	2	1	6	14	19
1_G10	69 female	0	49	35	33	15	7	16	3	2	7	20	20
1_G11	61 male	1	45	26	28	5	2	6	2	2	5	11	46
1_G12	73 female	0	39	28	28	14	9	14	6	3	8	19	24
1_H1	75 male	1	53	43	46	8	5	9	3	3	6	24	21
1_H2	60 male	1	38	28	29	11	8	17	14	1	4	10	15
1_H3	63 male	1	34	21	23	4	2	7	1	3	7	15	26
1_H4	65 female	0	25	21	20	10	5	9	5	2	6	14	22
1_H5	60 male	1	38	25	27	11	6	10	4	1	4	10	17
1_H6	68 female	0	48	38	39	7	4	8	2	2	6	16	22
1_H7	66 female	0	42	32	33	9	4	10	4	1	6	22	24
1_H8	67 female	0	44	31	31	10	4	11	4	1	4	12	18
1_H9	60 female	0	38	29	28	10	4	9	4	1	6	17	23
1_H10	61 female	0	54	43	45	9	4	10	2	3	11	22	21
1_H11	62 female	0	47	35	35	7	3	7	2	2	6	15	18
2_A2	28 male	1	77	56	60	5	0	10	2	2	4	11	11
2_A3	30 male	1	81	63	67	4	2	6	2	2	5	13	16
2_A5	27 male	1	75	55	58	7	3	7	3	1	3	12	12
2_A7	21 female	0	88	66	72	3	0	5	0	1	3	15	17
2_A8	31 male	1	65	48	48	7	4	9	2	1	4	13	14
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2_B1	29 female	0	61	40	45	6	1	17	1	3	8	23	24
2_B2	22 female	0	63	45	50	2	2	9	2	8	6	11	21
2_B3	27 male	1	78	56	63	6	3	6	2	1	3	11	15
2_B5	17 male	1	60	46	50	5	1	14	0	2	5	8	12
2_B8	16 male	1	82	69	73	2	1	5	1	1	3	12	14
2_B9	19 male	1	80	64	65	4	6	7	1	3	4	11	15
2_C1	19 female	0	84	76	81	4	2	7	1	2	5	13	16
2_C2	16 male	1	83	65	71	3	1	8	1	2	3	13	16
2_C3	15 female	0	83	74	78	3	2	5	1	1	4	14	17
2_C4	19 female	0	88	74	76	5	5	11	5	2	4	14	24
2_C6	18 female	0	82	60	65	7	7	9	6	1	6	13	14
2_C9	34 male	1	76	58	61	6	3	8	2	2	3	13	19
2_C10	34 male	1	69	52	55	5	2	5	2	2	4	14	17
2_C11	37 male	1	74	52	54	6	3	5	2	1	4	13	14
2_D1	38 male	1	63	41	42	7	3	6	3	2	4	12	13
2_D2	33 male	1	78	63	66	8	3	14	2	2	5	12	15
2_D5	37 female	0	64	41	44	4	2	6	1	1	8	17	20
2_D7	36 male	1	71	48	53	5	2	6	2	1	4	12	15
2_D8	34 male	1	81	65	67	6	4	6	1	1	4	14	15
2_D9	36 male	1	72	54	56	6	2	7	1	2	4	12	15
2_D10	36 female	0	74	55	57	6	2	7	2	2	5	12	15
2_D11	38 male	1	79	61	63	7	4	5	2	3	8	16	23
2_E1	40 male	1	70	55	59	7	3	6	2	1	5	15	17
2_E2	41 male	1	58	39	41	8	4	12	2	2	5	16	21
2_E4	42 male	1	60	43	50	7	3	4	2	2	6	13	16
2_E8	42 male	1	61	42	42	11	6	7	0	2	9	17	22

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2_F1	43 male	1	62	46	50	13	7	11	2	0	#NULL!	#NULL!	#NULL!
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2_F3	42 male	1	69	46	51	8	3	8	3	1	4	12	13
2_F4	41 male	1	67	50	54	8	4	8	2	2	6	16	19
2_F5	50 female	0	44	30	33	8	5	9	2	2	7	15	19
2_F6	51 female	0	59	42	46	9	5	10	2	0	7	23	21
2_F7	51 male	1	64	47	48	14	5	10	1	8	15	16	16
2_F8	51 male	1	56	38	40	9	3	13	2	1	5	16	18
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2_F12	51 male	1	81	63	77	15	4	14	2	0	#NULL!	#NULL!	#NULL!
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2_G2	51 male	1	60	43	46	12	4	13	2	1	5	13	17
2_G3	50 female	0	60	42	48	8	4	9	2	1	4	13	18
2_G4	52 female	0	62	43	45	13	10	14	8	3	7	17	21
2_G5	50 male	1	70	49	53	9	6	13	2	1	6	15	15
2_G6	51 male	1	53	39	39	3	2	11	1	2	7	18	18
2_G7	51 male	1	55	41	42	12	8	14	4	3	9	20	27
2_G9	62 female	0	48	34	38	10	5	11	3	3	6	16	23
2_G10	62 male	1	63	46	45	9	6	15	2	2	5	13	16
2_G12	62 male	1	48	31	33	11	7	11	3	1	1	27	31
2_H1	62 male	1	58	41	45	6	4	12	3	2	4	12	16
2_H2	62 female	0	52	39	41	11	6	13	2	2	5	13	21
2_H3	63 male	1	54	40	38	19	9	13	5	1	4	17	22
2_H4	62 female	0	57	42	44	9	4	12	2	1	3	13	16
2_H5	62 male	1	46	31	33	7	4	7	1	2	3	13	17
2_H6	62 male	1	63	47	44	9	8	9	1	1	5	16	20
2_H7	60 female	0	63	46	49	7	3	10	2	3	7	20	17
2_H8	60 male	1	46	32	34	4	2	9	1	2	6	14	16
2_H9	62 female	0	57	45	49	7	4	10	3	1	11	28	18
2_H10	63 male	1	36	32	40	13	6	10	4	3	5	14	17
2_H12	68 male	1	47	36	34	4	3	7	2	3	7	13	17
3_A1	18 male	1	86	73	78	4	3	9	2	3	5	12	15
3_A2	15 male	1	84	72	79	8	6	10	2	2	5	14	15
3_A3	19 female	0	84	69	74	4	3	8	2	3	5	13	16
3_A4	18 female	0	87	72	76	5	4	7	3	2	5	13	16
3_A5	11 male	1	89	76	79	5	3	9	1	2	4	9	10
3_A6	19 female	0	87	68	72	5	1	8	1	2	5	14	15
3_A7	17 female	0	87	69	72	7	13	13	4	3	5	15	16
3_A8	19 female	0	89	75	80	5	3	10	2	3	5	16	20
3_A9	15 female	0	87	68	77	6	5	15	4	4	4	12	26
3_A10	19 male	1	85	66	71	5	3	9	3	3	5	15	17
3_A11	18 female	0	90	75	80	5	2	8	2	2	5	13	16
3_A12	18 male	1	89	75	79	6	6	9	3	2	4	10	13
3_B2	16 male	1	90	79	82	4	3	7	1	3	6	14	16
3_B3	19 female	0	88	70	73	7	3	8	0	2	6	18	16
3_B4	19 female	0	88	73	88	6	4	10	2	2	6	15	21
3_B5	26 female	0	81	63	72	7	4	12	0	3	5	15	17
3_B7	22 female	0	84	67	73	4	3	9	1	2	4	12	14
3_B8	22 female	0	78	59	63	5	3	11	1	2	5	11	15
3_B9	29 female	0	76	62	63	7	2	8	2	3	6	18	23
3_B10	24 male	1	84	66	70	5	3	8	2	2	5	12	14
3_B11	26 female	0	82	68	71	5	2	7	2	2	6	14	16
3_C1	39 male	1	75	55	57	6	3	9	2	2	3	11	13

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3_C2	39 male	1	68	51	44	7	3	9	2	2	4	13	15
3_C3	40 male	1	72	55	58	5	2	11	1	2	4	13	16
3_C4	40 male	1	72	57	61	7	2	8	2	2	5	13	17
3_C5	39 male	1	64	44	49	6	4	8	2	2	5	11	14
3	40 male	1	66	49	51	8	2	8	4	3	6	12	17
3_C7	40 male	1	71	53	54	8	4	12	2	2	6	13	16
3_C8	40 male	1	#NULL!	#NULL!	#NULL!	14	13	27	3	1	4	12	16
3_C9	39 male	1	79	61	64	10	4	13	3	4	5	10	18
3_C10	37 male	1	67	51	55	8	4	11	2	3	5	14	17
3_C11	40 male	1	78	61	63	7	4	11	2	2	5	15	16
3_C12	41 male	1	71	54	56	9	5	10	2	4	5	15	17
3_D2	31 female	0	82	66	67	5	3	8	2	2	7	17	18
3_D3	32 female	0	73	52	56	5	2	9	2	2	5	12	20
3_D4	31 female	0	75	57	60	5	2	9	1	2	6	16	17
3_D5	30 female	0	80	58	65	5	2	8	2	2	7	26	34
3_D6	31 female	0	77	55	57	5	2	9	2	3	6	2	19
3_F6	56 male	1	39	31	30	9	4	11	3	2	6	14	17
3_F7	55 female	0	56	40	43	9	5	12	3	3	5	15	21
3_F11	56 female	0	56	40	43	11	5	12	3	4	6	17	22
3_G3	56 male	1	46	30	33	9	4	10	2	3	7	13	0
3_G4	56 female	0	57	47	49	9	7	12	3	4	6	16	7
3_G7	74 female	0	42	34	34	11	7	14	3	4	9	22	26
3_G9	75 male	1	41	33	35	9	5	14	3	3	6	13	20
3_G10	75 female	0	48	39	39	10	4	12	3	3	8	18	23
3_G11	75 male	1	53	37	39	11	7	12	3	2	6	15	24
3_G12	75 male	1	56	39	42	16	9	15	5	2	5	14	19
3_H4	74 female	0	54	41	44	14	8	15	4	4	7	13	26
3_H6	72 male	1	38	32	32	11	7	13	4	3	6	17	25
3_H12	70 male	1	46	35	36	19	11	19	6	3	5	15	22
4_A3	45 male	1	75	51	53	8	4	7	2	2	5	12	14
4_A4	45 male	1	73	59	63	9	5	11	3	2	6	15	18
4_A6	46 male	1	75	57	61	7	3	8	2	3	5	15	15
4_A7	50 male	1	52	36	39	7	3	7	3	2	5	6	26
4_A11	57 female	0	#NULL!	#NULL!	#NULL!	16	15	14	6	4	5	17	18
4_A12	61 male	1	53	40	42	8	4	9	4	2	6	16	22
4_B1	18 female	0	79	63	67	3	1	6	1	2	5	15	16
4_B2	18 male	1	87	72	76	4	2	6	1	1	3	8	14
4_B12	62 male	1	59	47	48	10	5	9	3	2	5	18	18
4_C1	18 female	0	79	63	67	3	2	6	1	2	4	11	16
4_C3	45 male	1	64	51	52	6	8	5	3	1	6	18	18
4_C4	48 male	1	71	53	54	9	4	6	3	3	5	16	16
4_C5	47 male	1	#NULL!	#NULL!	#NULL!	3	4	4	1	2	9	12	17
4_C6	47 male	1	65	47	49	7	4	9	2	3	12	12	27
4_C7	50 male	1	57	43	44	9	4	10	3	0	3	10	27
4_C8	55 female	0	50	33	34	7	3	7	3	0	10	19	20
4_C9	56 female	0	64	49	52	7	3	7	2	2	7	19	18
4_C10	57 female	0	55	41	42	9	4	9	2	2	5	16	19
4_C12	61 male	1	59	42	44	11	6	10	4	2	5	14	16
4_D1	17 female	0	87	75	79	4	2	6	1	1	5	14	18
4_D2	15 female	0	83	70	72	3	1	5	1	2	5	16	16
4_D3	45 male	1	70	51	54	7	3	8	2	2	4	15	19
4_D4	45 male	1	63	47	48	6	2	6	1	2	5	16	16
4_D5	46 male	1	40	28	30	5	3	7	2	2	10	19	22

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4_D6	46 male	1	58	43	44	7	3	7	2	2	4	14	14
4_D7	53 female	0	47	33	35	6	3	7	2	2	7	22	22
4_D8	54 male	1	59	44	46	5	3	6	2	2	5	13	15
4_D9	56 male	1	61	43	45	7	4	9	2	2	7	20	20
4_D10	56 female	0	64	51	54	9	7	10	4	2	7	18	20
4_D12	72 female	0	32	26	27	7	3	6	2	2	6	20	20
4_E1	13 male	1	84	73	77	5	4	7	2	1	4	13	14
4_E5	46 male	1	58	48	47	9	5	7	1	3	4	17	19
4_F8	54 male	1	59	46	50	6	3	9	1	1	4	17	14
4_F9	58 male	1	56	39	40	9	5	11	3	2	6	15	15
4_F12	73 female	0	41	35	36	19	7	15	6	2	7	24	24
4_G1	72 female	0	39	31	32	8	4	9	3	2	6	19	21
5_B3	12 male	1	89	80	84	3	3	6	1	1	3	12	12
5_B4	15 male	1	87	77	79	5	3	11	1	3	5	13	14
5_B5	15 male	1	90	76	80	2	2	6	1	1	5	13	13
5_B8	15 male	1	85	67	73	3	1	5	1	1	4	13	13
5_B11	31 male	1	70	49	51	7	2	8	2	1	4	17	16
5_B12	32 male	1	75	56	60	5	3	7	1	2	7	18	22
5_C1	24 male	1	78	63	67	4	2	9	2	3	4	15	16
5_C2	29 male	1	75	52	56	8	4	10	2	2	5	14	16
5_C3	25 female	0	74	60	63	4	3	7	3	2	5	15	18
5_C4	21 female	0	76	54	57	4	2	9	2	2	3	14	15
5_C5	29 male	1	86	66	62	5	2	7	1	1	4	14	13
5_C6	22 male	1	81	62	64	3	2	7	2	2	3	9	12
5_C7	22 male	1	83	66	72	2	1	6	1	2	3	14	11
5_C8	26 female	0	70	50	52	3	2	7	1	2	5	12	14
5_C9	20 female	0	82	66	69	4	1	7	1	2	5	11	14
5_C10	25 male	1	78	61	63	4	2	6	1	2	3	12	13
5_C11	21 male	1	82	64	70	5	2	8	2	2	6	15	15
5_C12	25 male	1	85	67	71	4	2	6	1	2	5	14	15
5_D1	21 female	0	79	60	64	5	2	9	2	3	6	16	22
5_D2	32 male	1	78	58	64	6	3	10	3	3	5	11	14
5_D3	34 male	1	73	55	59	6	3	10	2	3	7	17	18
5_D4	30 female	0	72	57	59	6	2	10	2	3	6	14	17
5_D5	32 male	1	79	58	63	5	2	7	1	2	4	13	19
5_D6	39 female	0	74	54	58	5	3	7	2	2	6	17	19
5_G1	66 male	1	57	43	42	14	7	14	4	2	6	16	19
5_G7	66 male	1	49	35	35	13	8	14	3	2	5	16	26
8_H1	69 male	1	51	39	36	13	7	15	4	4	6	17	19
8_H2	70 male	1	58	43	44	11	6	11	5	1	5	18	24
8_H3	67 male	1	55	43	44	9	5	10	3	2	6	17	21
8_H4	70 male	1	60	46	47	12	7	11	4	3	7	19	21
8_H5	70 male	1	56	41	43	13	6	12	4	2	5	17	19
8_H6	68 male	1	48	35	35	15	10	15	4	2	5	17	16
8_H7	70 male	1	50	36	36	17	10	14	5	2	4	16	18
8_H8	67 male	1	56	44	45	6	3	6	2	2	5	16	19
8_H9	70 male	1	48	40	40	11	6	12	3	2	5	16	23
8_H10	65 male	1	49	36	37	8	4	8	2	2	6	16	20
8_H11	2 male	1	86	85	88	3	1	5	1	1	3	11	12
9_A3	2 male	1	90	89	92	3	1	6	1	1	4	11	13
9_A4	2 male	1	92	90	92	3	2	6	1	2	4	11	12
9_A5	3 female	0	89	89	91	3	2	7	2	2	4	10	13
9_A6	2 female	0	91	89	90	2	2	10	1	2	5	13	13
9_A8	2 female	0	89	89	92	3	2	7	1	2	4	12	10

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Suplementary Table S4. DNA methylation values obtained for the 120 sample testing set in the 32 analysed C markers included in the model.

sample	age	sex	sex_bin	C1ORF132_C1	C1ORF132_C2	C1ORF132_C3	KLF_C1	KLF14_C2	KLF14_C3	KLF14_C4	elovl2_c1	
1_A2	19	female	0	84	71	80	5	2	7	2	30	
1_A4	19	male	1	86	67	71	4	2	5	1	27	
1_A6	27	male	1	71	54	55	3	2	5	1	35	
1_A7	27	female	0	84	64	71	4	2	5	1	30	
1_A8	28	female	0	62	48	53	2	1	10	1	28	
1_A9	28	male	1	81	59	64	7	3	6	3	35	
1_A10	20	female	0	87	72	75	6	3	7	2	16	
1_B1	27	male	1	77	55	60	4	2	6	2	39	
1_B2	23	female	0	87	69	75	5	2	8	2	22	
1_B7	22	male	1	77	61	64	6	2	5	2	27	
1_C2	26	male	1	69	51	54	3	2	6	1	28	
1_C4	32	female	0	68	50	53	7	2	6	2	31	
1_C7	33	female	0	72	47	52	4	3	6	1	30	
1_C8	39	male	1	76	60	64	5	2	7	2	47	
1_C9	35	female	0	73	48	54	5	2	8	1	31	
1_D6	35	male	1	63	47	50	5	2	6	2	34	
1_D9	38	male	1	76	55	59	7	3	10	3	37	
1_F4	44	female	0	68	49	54	5	2	8	1	34	
2_A4	22	female	0	78	65	67	3	0	6	0	28	
2_A6	30	male	1	81	54	63	8	4	7	4	30	
2_A10	27	female	0	70	51	60	4	3	18	3	41	
2_A11	26	female	0	80	74	75	7	5	10	2	26	
2_B4	28	male	1	74	64	62	8	2	7	0	31	
2_B6	17	female	0	83	72	74	2	1	7	1	24	
2_B7	14	male	1	84	71	72	5	2	5	0	19	
2_B10	19	male	1	85	72	73	6	2	10	2	29	
2_B11	18	male	1	80	62	66	4	1	10	1	21	
2_B12	19	female	0	83	75	80	3	1	8	0	18	
2_C5	18	female	0	85	65	82	5	3	4	3	13	
2_C7	18	female	0	87	75	81	8	4	9	4	32	
2_C8	19	female	0	90	70	75	3	3	8	2	15	
2_C12	37	female	0	66	47	50	5	6	9	4	26	
2_D4	36	male	1	69	51	53	5	2	5	1	31	
2_D6	38	male	1	72	56	55	5	2	5	3	38	
2_D12	35	female	0	60	45	48	9	7	11	3	34	
2_E7	40	male	1	71	48	53	5	2	5	1	45	
2_E9	44	male	1	62	42	45	10	4	7	3	36	
2_E10	43	male	1	52	38	41	12	6	15	4	36	
2_E11	41	male	1	70	54	57	11	3	10	0	45	
2_E12	38	male	1	74	52	56	7	5	10	2	31	
2_F11	51	female	0	71	47	60	12	8	13	3	41	
2_G11	62	female	0	49	36	38	11	7	14	3	46	
2_H11	63	male	1	50	42	41	9	4	11	2	41	
3_B1	18	female	0	88	75	79	5	2	9	2	32	
3_B6	29	female	0	77	59	66	7	3	12	3	35	
3_D1	33	female	0	74	57	61	6	3	8	2	31	
3_E12	50	female	0	61	46	47	13	7	13	4	34	

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3_F1	55 female	0	63	44	48	13	6	13	4	39
3_F10	56 female	0	54	45	47	8	6	11	3	46
3_G5	56 female	0	58	48	51	12	6	11	3	42
3_G6	57 male	1	50	40	41	8	4	11	2	52
3_G8	74 female	0	48	38	36	16	9	16	5	49
3_H1	71 female	0	52	38	41	11	6	11	3	54
3_H2	73 female	0	44	33	35	11	6	12	4	53
3_H3	73 male	1	60	45	46	8	4	9	3	45
3_H5	72 male	1	52	38	42	8	4	9	3	58
3_H7	70 female	0	53	38	38	20	9	17	5	43
3_H8	72 female	0	40	32	32	13	7	13	4	48
3_H9	70 male	1	50	38	39	16	7	17	3	42
3_H10	75 female	0	61	46	48	13	7	13	4	51
4_E2	18 female	0	88	74	78	4	2	6	1	23
4_E3	45 male	1	59	39	40	6	3	7	2	39
4_E4	46 male	1	58	40	38	14	5	11	4	45
4_E6	46 male	1	59	42	43	11	6	12	4	36
4_E7	53 female	0	54	41	42	7	2	6	2	40
4_E9	55 male	1	53	40	40	5	3	9	2	48
4_E10	56 female	0	44	33	34	7	3	9	1	40
4_E11	62 female	0	40	27	29	9	4	7	3	47
4_E12	72 female	0	40	31	32	10	6	10	3	46
4_F1	19 female	0	90	74	79	4	2	6	1	22
4_F2	16 female	0	83	70	72	4	1	5	1	26
4_F3	46 male	1	70	54	54	4	3	6	2	36
4_F4	45 male	1	67	54	54	9	3	9	2	38
4_F6	46 male	1	53	43	46	6	2	8	2	33
4_F7	52 female	0	68	51	54	10	4	8	3	40
4_F10	57 female	0	43	32	34	9	5	8	3	39
4_F11	63 female	0	52	41	40	10	6	13	3	40
4_G2	73 female	0	47	36	38	9	5	8	2	40
4_G3	70 female	0	49	38	39	6	3	8	2	51
4_G4	74 male	1	27	23	24	10	5	9	4	58
4_G5	73 male	1	47	29	34	15	7	11	3	40
4_G6	70 male	1	49	37	38	12	7	6	3	43
4_G7	74 female	0	48	39	41	13	6	10	4	50
4_G8	74 female	0	57	46	48	13	6	13	4	51
4_G9	75 female	0	55	42	45	12	5	13	4	43
4_G11	72 male	1	50	41	42	12	6	11	3	49
4_G12	70 male	1	50	39	40	14	6	16	4	45
4_H1	57 male	1	55	40	44	9	5	8	3	42
4_H2	57 male	1	47	34	35	10	5	10	3	39
4_H3	60 male	1	57	41	42	13	7	13	3	48
4_H5	57 male	1	60	40	42	8	5	7	3	51
4_H12	52 male	1	49	36	36	6	3	8	2	43
5_A3	19 female	0	85	70	74	5	2	6	2	25
5_A4	19 female	0	84	65	69	6	3	10	3	22
5_A6	19 female	0	80	63	66	3	2	8	1	30
5_A7	19 male	1	85	69	72	5	2	7	2	24
5_A8	18 female	0	73	59	66	2	2	4	0	23
5_A9	19 female	0	83	66	70	4	1	8	2	28
5_A10	19 female	0	85	65	72	7	3	10	1	31
5_A11	17 male	1	83	63	67	3	1	5	1	23
5_A12	18 female	0	86	73	78	3	1	5	1	22
5_B2	17 female	0	85	73	76	4	2	7	2	22

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5_B6	19 male	1	84	69	73	4	2	8	1	25
5_B7	19 female	0	83	73	74	3	1	6	1	23
5_B9	19 female	0	85	70	73	4	2	4	1	26
5_B10	14 female	0	83	69	71	3	1	5	1	24
5_D7	34 female	0	73	56	57	5	2	7	2	30
5_D10	25 female	0	77	57	60	5	2	6	1	30
5_D11	32 male	1	78	59	63	6	3	8	1	29
5_E8	41 male	1	63	45	47	9	4	8	3	37
5_E9	44 male	1	56	47	46	11	6	7	2	41
5_E11	41 male	1	66	52	55	7	6	5	1	34
5_F1	41 male	1	65	46	48	5	2	7	2	35
5_F3	42 male	1	56	40	41	9	6	10	3	43
5_F5	48 male	1	52	40	41	5	3	7	2	50
5_G2	66 male	1	58	43	41	15	10	13	5	44
5_G4	63 male	1	67	50	50	15	7	14	4	38
5_G5	62 male	1	65	53	51	9	4	9	2	48
9_A2	3 male	1	87	88	87	4	2	7	1	16
9_A9	2 female	0	89	93	92	2	1	5	1	11

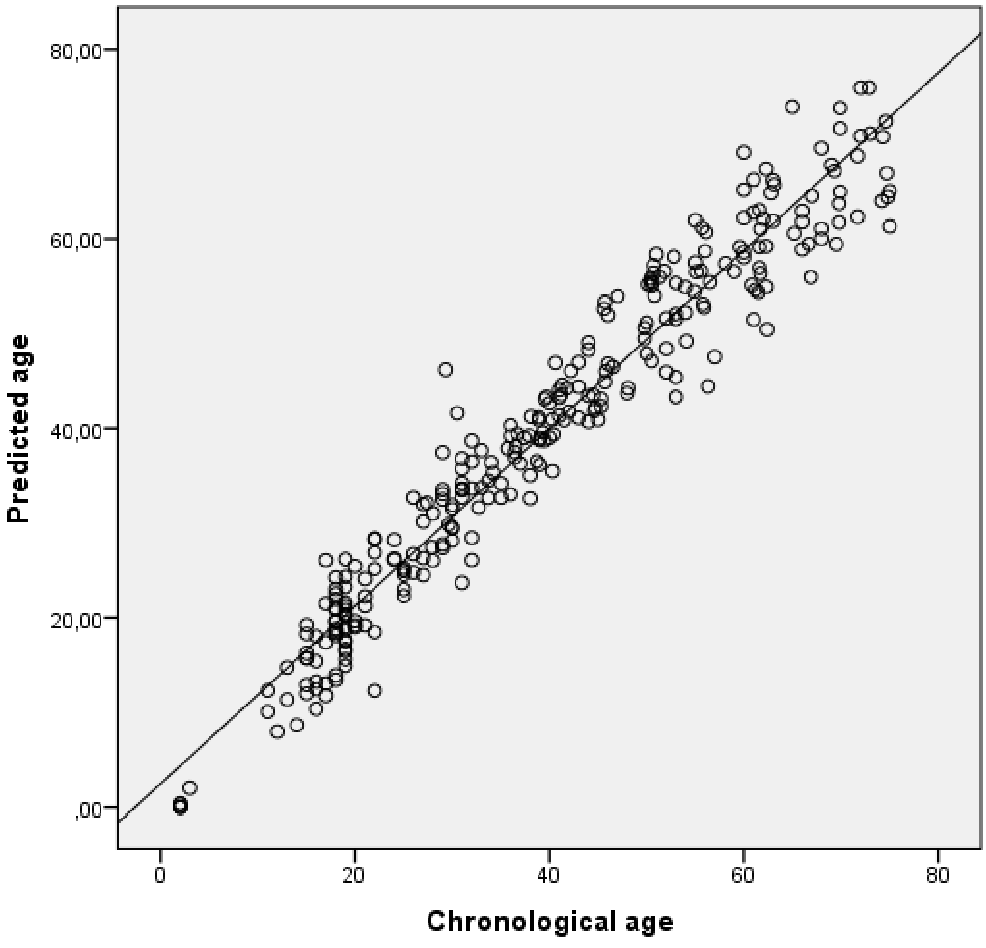


Fig1 .

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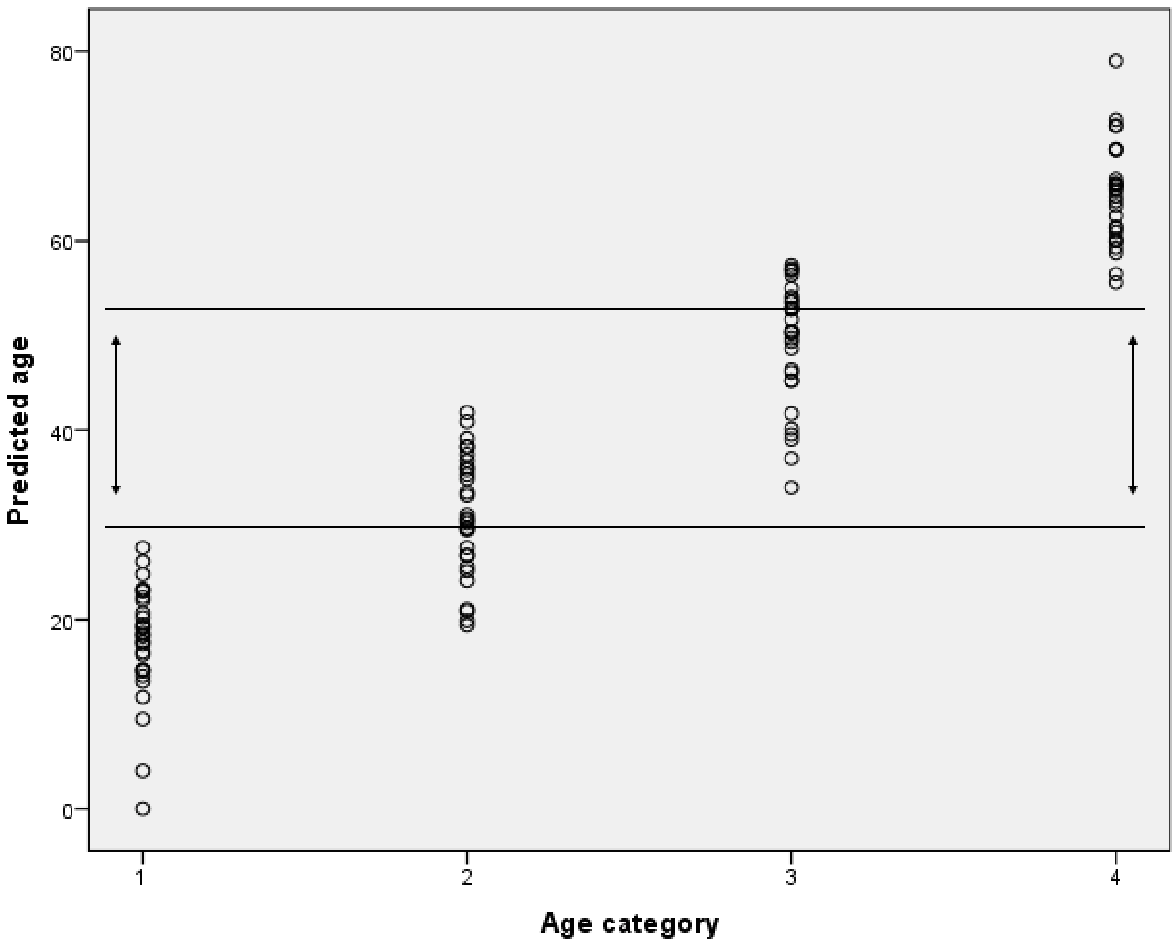


Fig2 .

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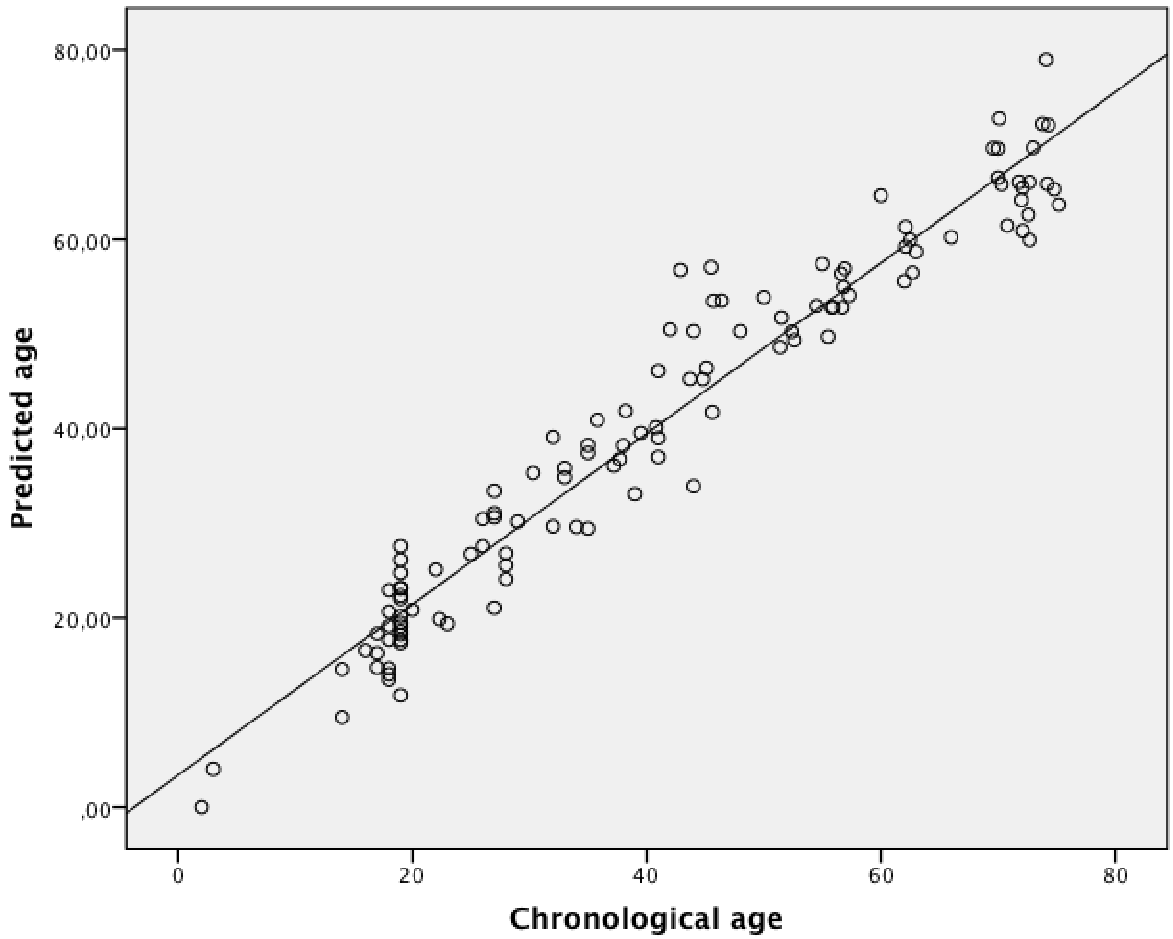
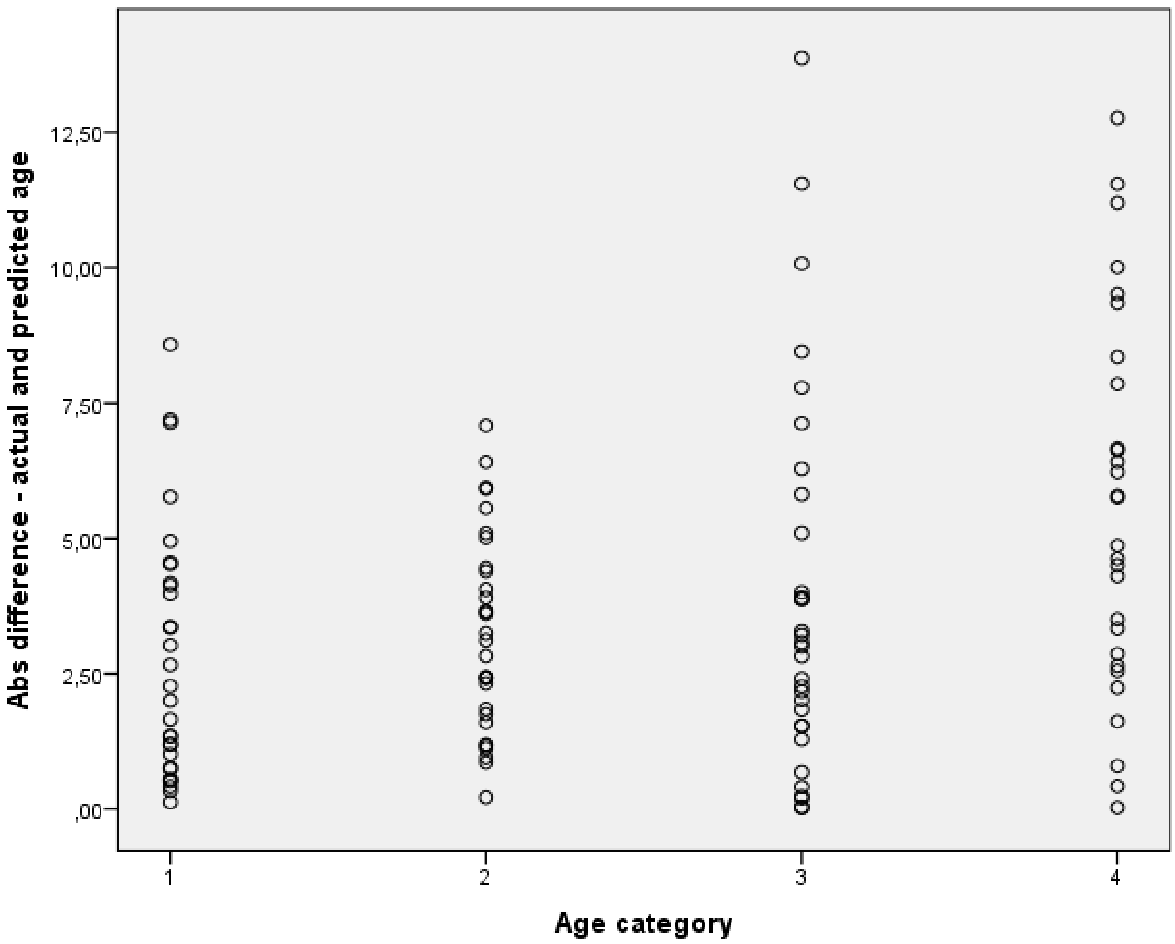


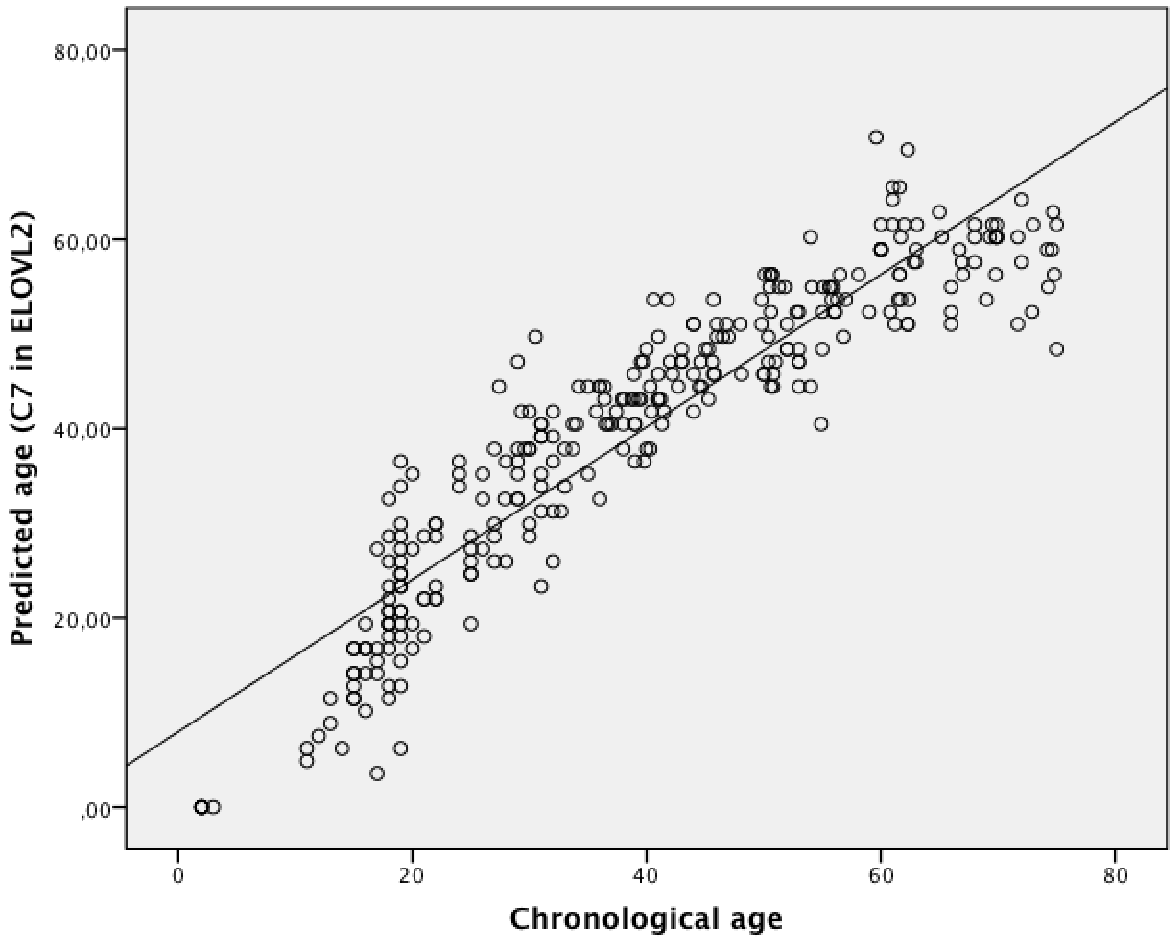
Fig3_revision .

ACCEPTED MANUSCRIPT



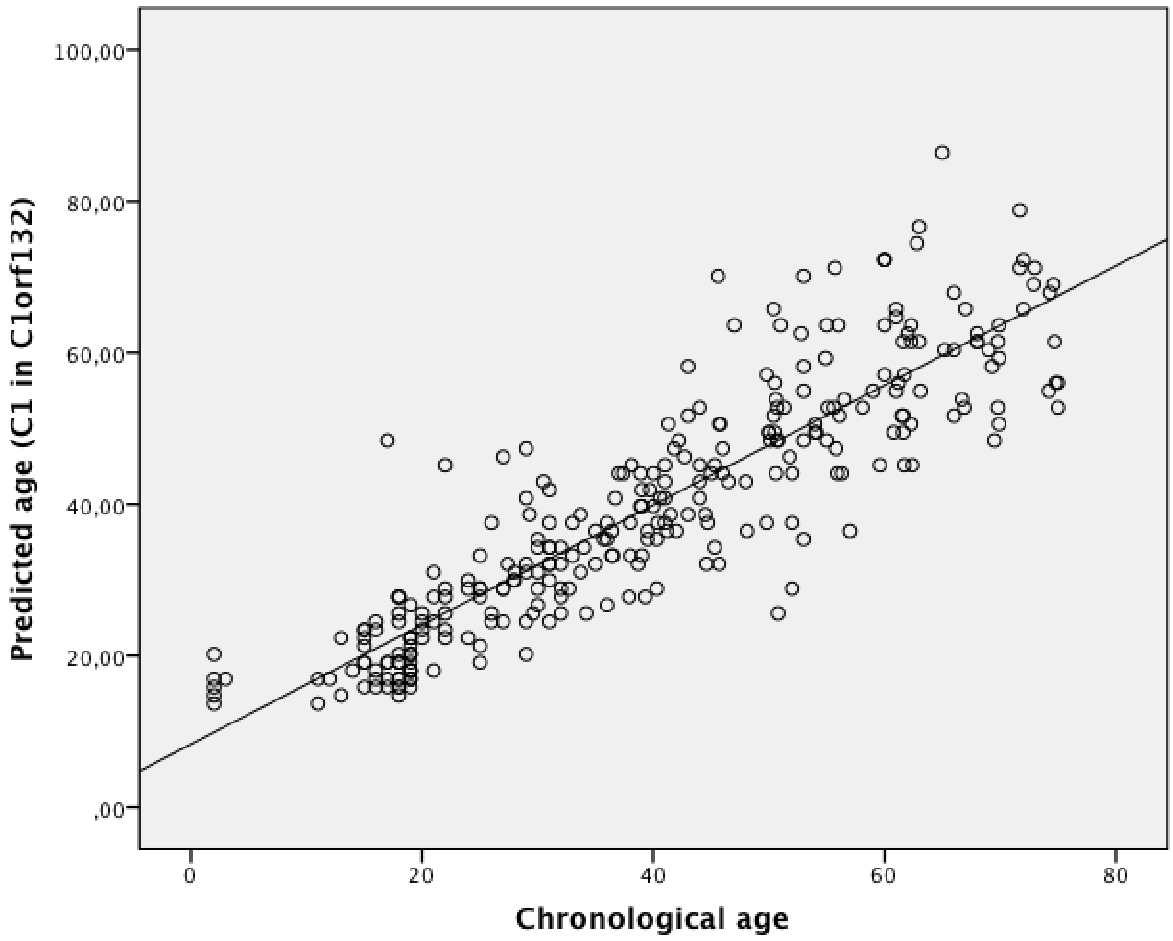
Supplementary Fig S1 .

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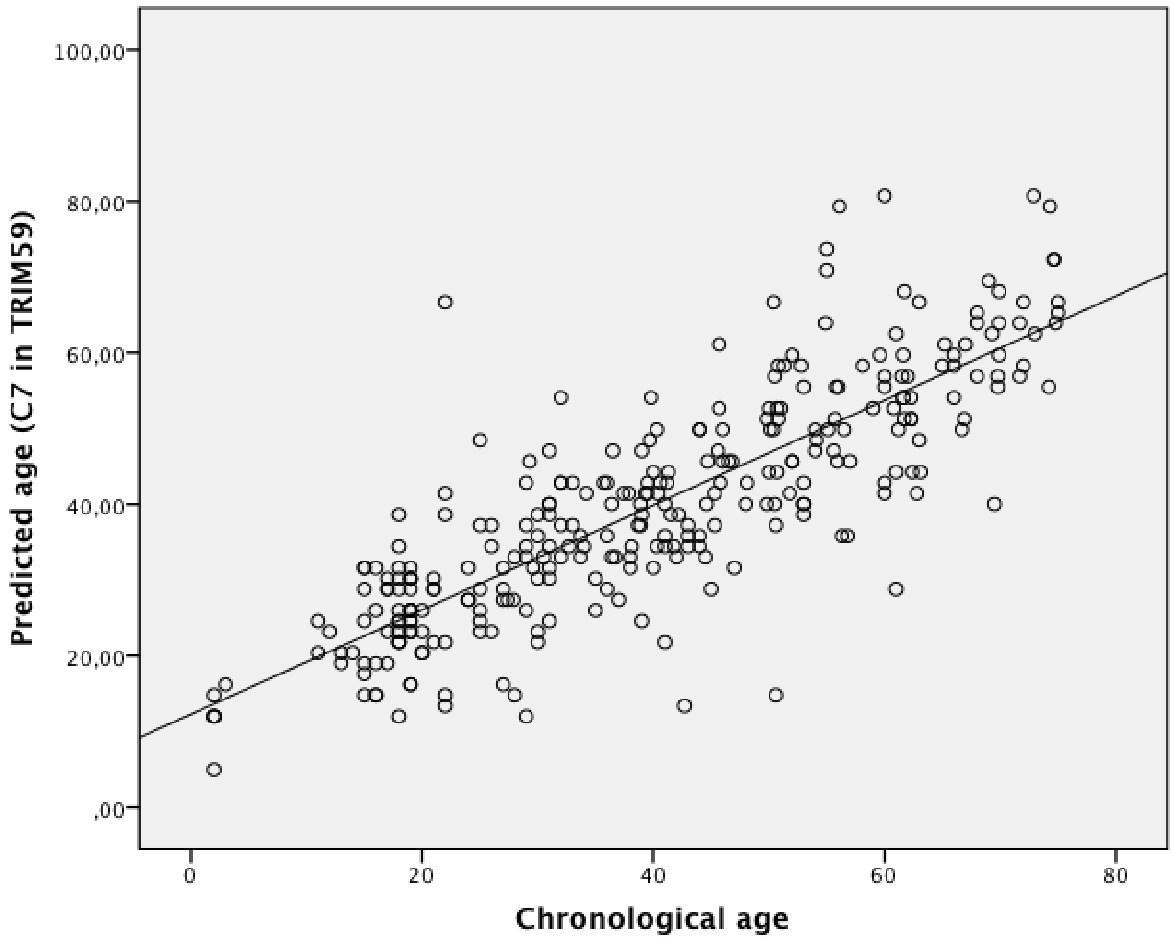
Supplementary Fig2_revision .

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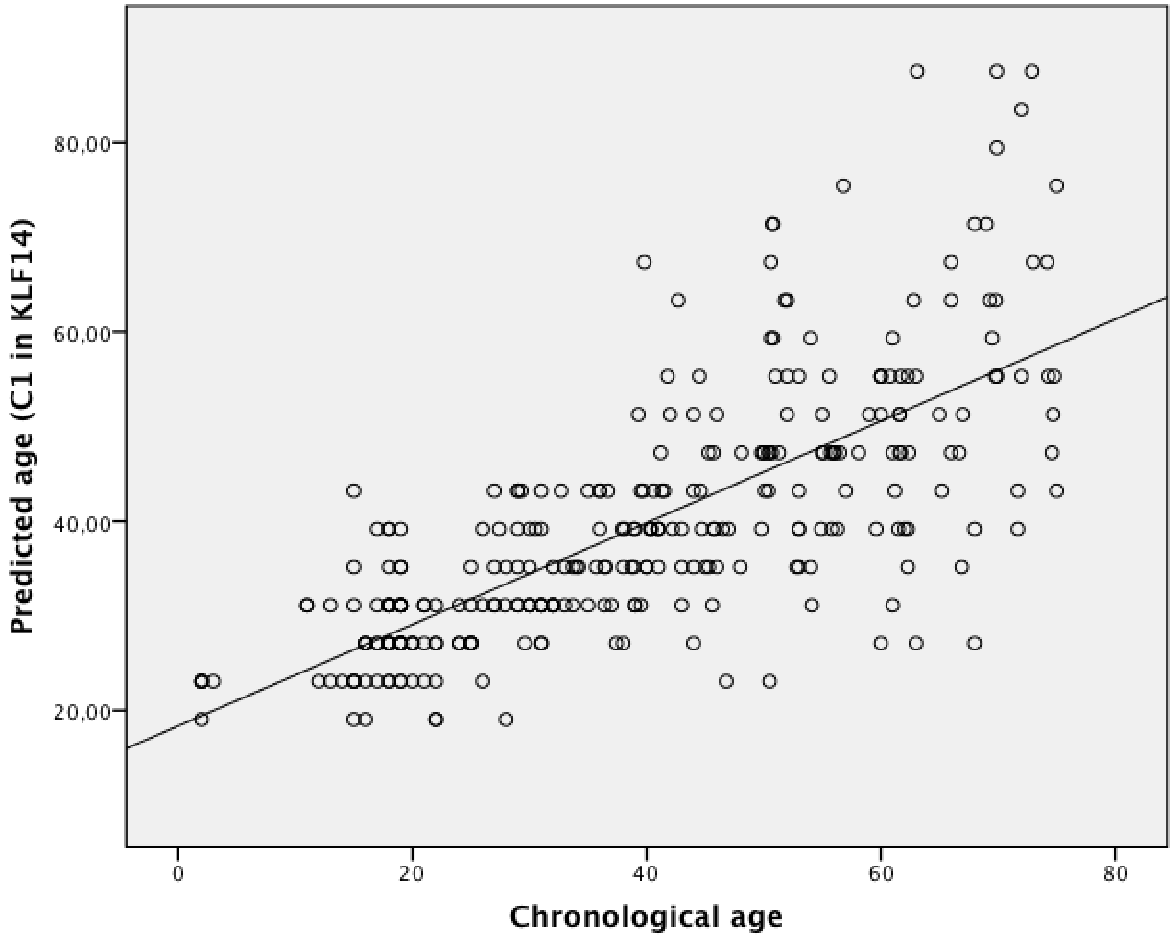
Supplementary Fig3_revision .

ACCEPTED MANUSCRIPT



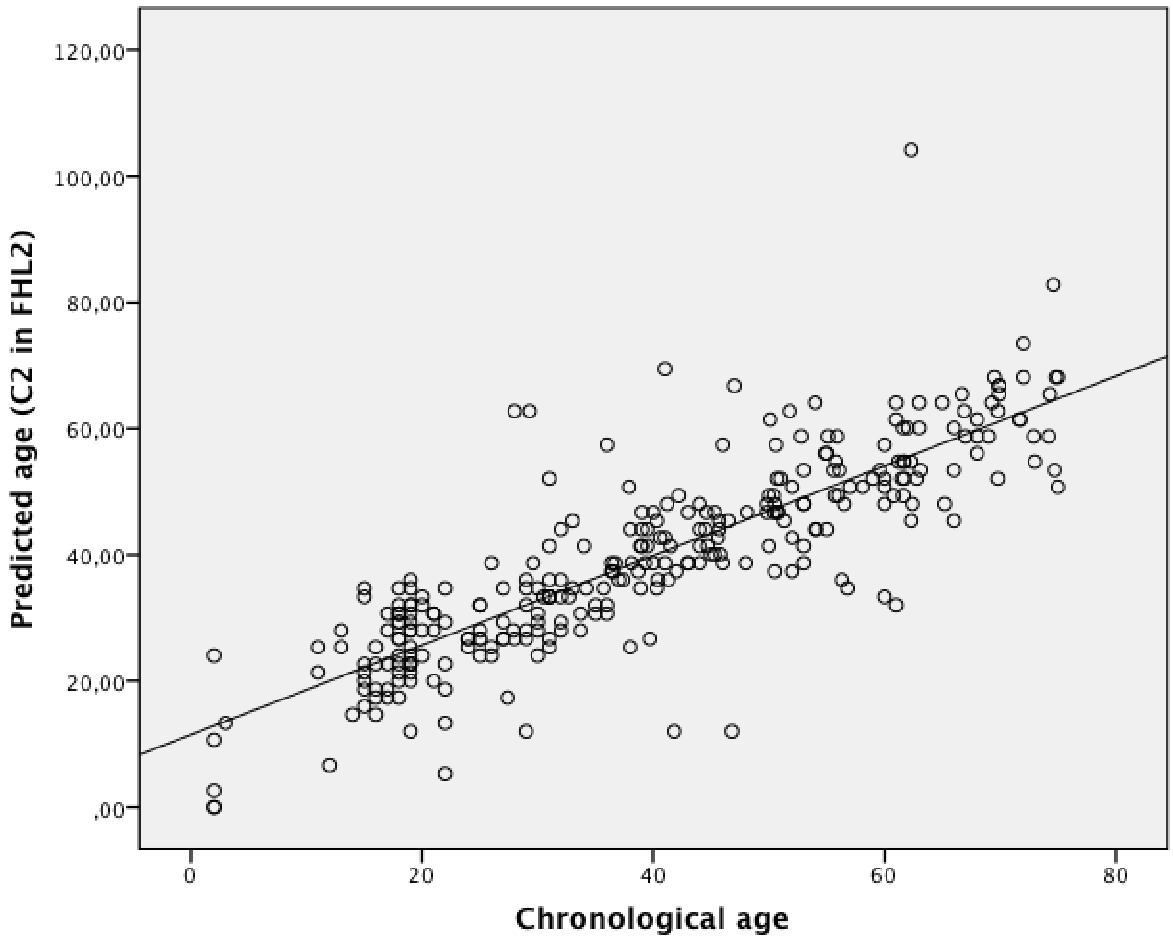
Supplementary Fig4_revision .

ACCEPTED MANUSCRIPT



Supplementary Fig5_revision .

ACCEPTED MANUSCRIPT



Supplementary Fig6_revision .